

WaveKitchen

A cooking game where every kitchen action is a signal-processing operation

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1 · Motivation and approach

[Presenter]

2 · Theory at every station

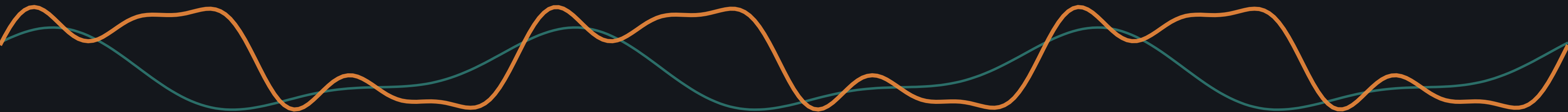
[Presenter]

**3 · Architecture and
implementation**

[Presenter]

4 · Results and live demo

[Presenter]



THE PROBLEM

Signals and Systems is taught as equations you cannot see or hear

Convolution

is a formula on the board, not
something you can hear.

Aliasing

is a warning in the textbook, not a
mistake you make.

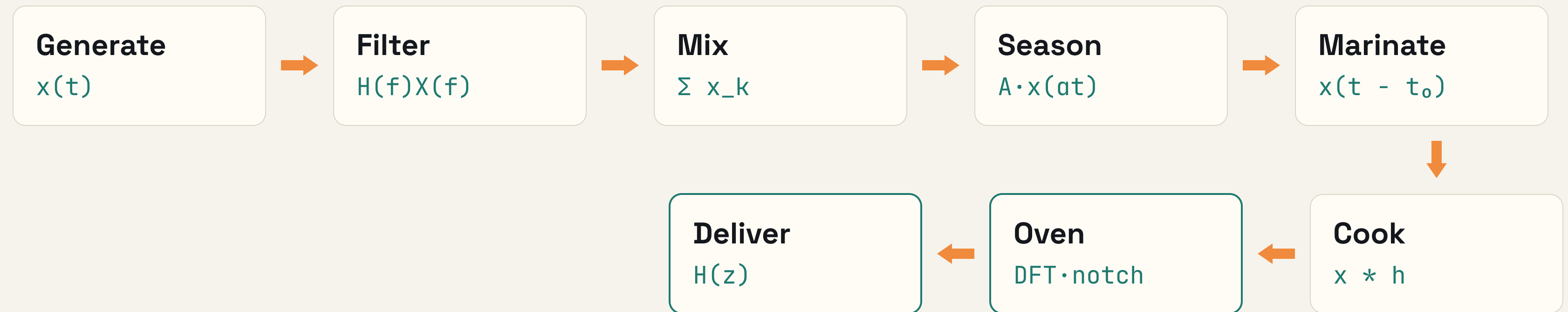
Poles and zeros

are dots on a plane, not a filter you
tune.

Our goal: give every concept a visible, audible consequence, and a score that tells you whether you got it right.

APPROACH

One dish, one signal chain



A run: pick one of **5 recipes** in 3 tiers and a difficulty (Easy 7:00 down to MasterChef 2:00), then match a hidden **reference dish**: the same chain on clean ingredients with the recipe's ideal settings.

■ finishing stations: their output is the served dish

Wash, mix, season, marinate

Washing

Low-pass filtering

$$Y(f) = H(f) \cdot X(f)$$

Noise arrives added, $x + n$.
Too low a cutoff cuts the ingredient itself.

Mixing

Superposition

$$y = (1/\sqrt{K}) \cdot \sum x_k$$

The bowl is linear; $1/\sqrt{K}$ keeps loudness comparable.
Curves add as 2-D paths.

Seasoning

Amplitude and time scaling

$$\begin{aligned} y &= A \cdot x(\alpha t) \\ \leftrightarrow (A/|\alpha|) \cdot X(f/\alpha) \end{aligned}$$

$\alpha > 1$ compresses time and moves energy up in frequency.

Marinating

Time shift

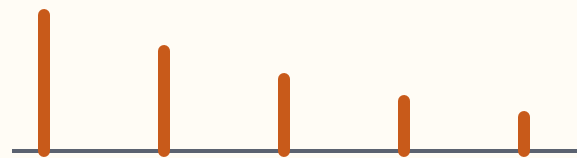
$$\begin{aligned} y &= x(t - t_o) \\ \leftrightarrow X(f) \cdot e^{-j2\pi f t_o} \end{aligned}$$

Ingredients are textbook signals: square and triangle waves (odd harmonics falling as $1/n$ and $1/n^2$), sums of sines, 2-D parametric curves and one real 44.1 kHz recording.

Cooking is convolution

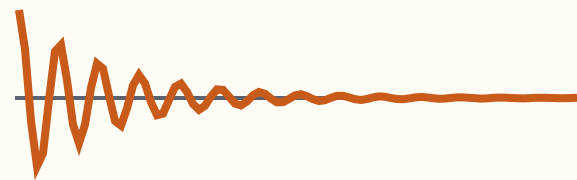
$$y[n] = \sum x[k] \cdot h[n - k]$$

An LTI system is fully described by its impulse response h . Each appliance is a different h ; the player slides h across x and watches the output build up.



Grill

A comb of echoes



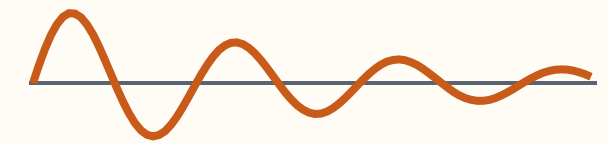
Fry

A bright, fast-decaying burst



Bake

A smooth long tail: a low-pass



Boil

A damped resonator

Kernels: 96 taps with $\sum |h| = 1$, so cooking reshapes the dish without changing its level.

Finishing: the oven and the cart

Precision Oven

Sampling · DFT · notch · inverse DFT

1. Sample at $f_s \geq 2 \cdot f_{\max}$, or the dish aliases
2. DFT: bins at $f_k = k \cdot f_s / N$
3. Spot the burnt overtone between two harmonics (burger: 18 Hz, between 16 and 20)
4. Notch it: gain $1 - 0.98 / (1 + (\Delta f / 1.5)^2)$
5. Inverse DFT: the rebuilt dish is served

System Delivery

Poles, zeros, stability, aliased sensing

$$H(z) = g \cdot (z - e^{j\omega_0})(z - e^{-j\omega_0}) / (z - 0.85e^{j\omega_0})(z - 0.85e^{-j\omega_0})$$

- Road tone: burger 2600 Hz at 8 kHz, $\omega_0 \approx 2.04$ rad
- Zeros on the unit circle at $\pm\omega_0$ remove it; poles at 0.85 keep the notch narrow
- A pole with $|p| \geq 1$ is unstable: the output grows
- A 1.2 kHz sensor reads 2600 Hz as 200 Hz: a notch there misses

The browser plays. The server judges.

Browser: plays

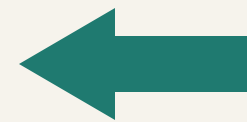
React 19 · TanStack Router/Start · TypeScript · Vite

- The DSP pipeline for every station
- Graphs and Web Audio playback
- Run state in localStorage
- Works offline with a local score

JSON over /api



settings: bowl, cutoffs,
dials, appliance, notch,
poles



scores, stages, ranks

Server: judges

FastAPI · SQLAlchemy · SQLite · NumPy/SciPy

- Seeded dirty ingredients per session
- Rebuilds and scores the dish
- Points, ranks and tier unlocks
- The leaderboard

The browser sends settings, never scores. The server recreates the dirty ingredients from the session seed, runs the same pipeline as the reference, claims each dish once and times the bonus on its own clock.

IMPLEMENTATION

How a dish becomes a score

Dish match (server)

$$m = 0.40 \cdot \text{spec} + 0.35 \cdot \text{corr} + 0.25 \cdot \text{SNR}$$

dB spectra compared over the 40 dB below the target's peak, correlation, SNR scored from -5 to 25 dB; both dishes on the reference scale, so the seasoning level counts

Final score

$$\text{raw} = m \cdot (0.72 + 0.28 \cdot \text{prep}/100)$$

$$\text{score} = 100 \cdot (\text{raw}/100)^\gamma$$

Washing scales the dish, so it can't carry a ruined one; $\gamma = 1 / \text{recipe tolerance}$; stars at 92 / 80 / 66 / 50 / 32

Overall score

$$\text{total} = \text{score} \cdot 10 \cdot d + \min(2 \cdot s, 300) \cdot d$$

$d = 0.8 / 1.0 / 1.25 / 1.5$ from Easy to MasterChef, s = seconds left; the leaderboard ranks this

Live match in each lab

$$0.5 \cdot \max(0, \rho) + 0.5 \cdot (1 - \text{RMSE}/\text{range})$$

ρ = Pearson correlation with the target; instant feedback while you play

Decisions and features

Hard problems we solved

- **Cart started cold:** a steady-state response cut the start-up error from about 100 % of the peak to under 5 %
- **Oven lost content above $f_s/2$:** the equaliser now acts on the full dish, so a perfect finish keeps 99–100 %
- **Normalising hid seasoning mistakes:** every comparison uses one common scale
- **Score page and leaderboard disagreed:** one overall score, computed and stored by the server

What the game offers

- 5 recipes, 3 tiers and 4 difficulties, each with a timer, a start countdown and pause
- Accounts, ranks, tier unlocks and a seasonal leaderboard: top 5, then See all
- Every signal can be heard; cooking sounds play while you convolve
- Signal Playground: 10 labs, including the oven, the z-plane, tasting and a full recipe chain
- Works offline with a local score

RESULTS

Precision pays, mistakes cost

100%

Perfect cooking matches the reference, on all 5 recipes

99–100%

Final dish after a perfect oven and cart, against the convolved dish

45–61%

Serving with the burnt overtone left in

34–39%

Marinating 20 % too long

Server case, all 5 recipes	Dish score
Perfect finish	93–98
Notch left off	17–36
Cart aimed at the aliased reading	9–23
Unstable cart	under 5

Tested end to end

187 frontend tests (Vitest)

108 backend tests (pytest)

0 type or lint errors

LIVE DEMO

One burger, start to finish

1. Burger on Medium: wash, mix, season, marinate, grill
2. Oven: safe f_s , notch at 18 Hz
3. Cart: notch at ω_0 , not the aliased reading
4. Score and leaderboard

KNOWN LIMITATIONS

- Browser and server build ingredients differently: same parameters, different samples
- The live match is mostly shape-based; the server score is stricter
- Pause time still counts toward the server's time bonus

Thank you

Questions?

Every team member can take questions on any part of the project.